BASSEM GHORBEL

+1 (970) 690-5258 ⋄ Fort Collins, CO

bassem@colostate.edu ⋄ LinkedIn ⋄ Google Scholar

SUMMARY

Computer Science PhD researcher specializing in formal methods, verification, and algorithm design for cyber-physical systems. Experienced in developing scalable monitoring and verification algorithms for temporal-logic specifications. Strong background in C++, Python, mathematical reasoning, and algorithmic problem solving.

EDUCATION

Colorado State University, Fort Collins, CO — Aug 2021 – Present
Ph.D. in Computer Science – Formal Methods — Expected Dec 2026
Robert B. France Fellowship Award Recipient, 2025

Ecole Polytechnique de Tunisie, Tunis, Tunisia — Sep 2018 – Jun 2021
Diplôme d'Ingénieur – Signals and Systems Specialization

IPEIS, Sfax, Tunisia — Sep 2016 – Jun 2018
Classes Préparatoires (Mathematics & Physics) – National Engineering Exam Rank: 57/2000

RESEARCH EXPERIENCE

Colorado State University, Fort Collins, CO — Aug 2021 – Present
Ph.D. Researcher – Formal Methods, Verification, and Cyber-Physical Systems

- Developed formal specification and quantitative robustness frameworks for temporal logics with freeze quantifiers, extending MTL/STL to express value-dependent and history-dependent behaviors in cyber-physical systems.
- Designed linear-time and quasi-linear runtime monitoring algorithms for temporal-logic specifications, improving the scalability of verification through dynamic programming and heuristic techniques.
- Developed mathematical proofs establishing error bounds between discrete- and continuous-time robustness under Lipschitz continuity assumptions, providing guarantees for sampled-data monitoring.
- Implemented and benchmarked verification algorithms in C++ and MATLAB, demonstrating scalable monitoring on large execution traces.

SELECTED PUBLICATIONS

- B. Ghorbel, V. S. Prabhu. "Fast Robust Monitoring for Signal Temporal Logic with Value Freezing Operators (STL*)." MEMOCODE 2024. ***Best Paper Award***.
- B. Ghorbel, V. S. Prabhu. "Fast and Scalable Monitoring for Value-Freeze Operator Augmented Signal Temporal Logic." HSCC 2024.
- B. Ghorbel, V. S. Prabhu. "Quantitative Robustness for Signal Temporal Logic With Time-Freeze Quantifiers." IEEE TCAD 2023.
- B. Ghorbel, V. S. Prabhu. "Linear Time Monitoring for One Variable TPTL." HSCC 2022.

TECHNICAL SKILLS

Programming	C++, Python, MATLAB, C
Algorithms	Algorithm Design & Analysis, Data Structures, Optimization, Complexity Analysis
Systems	Linux, Bash, Git
Other	SQL, Verilog, NumPy, Pandas
Languages	Arabic (Native), French (Bilingual), English (Fluent)

TEACHING EXPERIENCE

Teaching Assistant, Courses: Software Development with C++, Operating Systems, Foundations of Cyber-Physical Systems, Algorithms – Theory and Practice, Discrete Structures and their Applications.